Ram Calculus and Local Smoothing for the Euclidean Wave Equation

Quaternionic Structure, Auxiliary Dirichlet Problems, and Residual-Certified Reformulations

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Abstract

We investigate the October 25, 2026 OpenAI manuscript on Euclidean wave local smoothing, using the supplied PDF and its matching versioned source. The central auxiliary equation is a quaternionic Moore-determinant Dirichlet problem with a rank-one background, not another wave equation and not a constant-coefficient Ram partial differential operator. The corrected quaternionic Ram Master Transform of TR60, its revised foundational treatment, and the subsequent constrained-channel, frame, fractional-operator, and nonlinear RMT extensions provide relevant representation and validation tools. We give exact null-coordinate and Gaussian-semigroup reformulations, derive boundary-aware cofactor Newton equations, and prove residual-to-solution and residual-to-energy estimates under explicit admissibility assumptions. An off-solution energy identity isolates the product of determinant residual and spacetime Schur complement that an approximate solver must control. Further results quantify rank-one background asymptotics, planar disk inversion, coherent-current reconstruction, frequency-dependent frame errors, and dispersion fidelity. Worked models and deterministic numerical checks accompany the proofs. A bounded audit reproduces the principal algebraic cancellation and checks the main comparison directions, but does not constitute independent certification of every analytic estimate in the source. No improved unrestricted local-smoothing exponent or optimal logarithmic loss is proved here. The demonstrated contributions are precise equivalent formulations, conditional error certificates, and explicit obstructions to invalid inversion and approximation claims; sharper bounds and full computational validation remain research proposals.

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1 Scope, provenance, and the meaning of a proved contribution

The object of study is *Local smoothing for the Euclidean wave equation*, dated October 25, 2026 [1]. The supplied PDF has 59 pages. Its Git blob identifier is

`0472605895a485dd0441ada1257485d325f16f55.`

It matches the public PDF at repository commit

`42907c9e999cc00d6abe7a50bfc1023250edd1b9.`

The mathematical source files at that commit were retained locally for comparison. Overlines, quaternionic multiplication order, and boundary terms were checked in the LaTeX source rather than inferred solely from PDF text extraction.

The repository states that its manuscripts have different verification statuses and that unformalized results can contain issues [2]. Accordingly, the source's theorems are *claims of the source* unless the relevant argument is independently supplied here. This report is not a formal verification, a community acceptance statement, or a priority claim. The literature checks are dated October 26, 2026.

Three levels of assertions will be kept distinct:

1. **Exact identities and proved conditional results.** These have proofs below and precisely specified domains, boundary conditions, positivity margins, and norms.
2. **Source-dependent conclusions.** The critical smoothing estimate and its far-reaching consequences are reported with their dependencies. Recasting their proof does not make those dependencies disappear.
3. **Proposals.** Numerical speedups, sharper unrestricted logarithms, more general geometries, and relaxed nonlinear ellipticity assumptions are not presented as established results.

The corpus investigation was broad but targeted: revised operator and quaternionic reports, the four RMT-extension reports, relevant foundational volumes, prior wave and restriction analyses, determinant reports, compatible panels, and stable frames were inspected. This does not mean that every page of the very large corpus was independently verified. Confidential local sources were used locally; their text was not submitted in public web searches.

2 The exact wave problem and its quantifiers

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} e^{-\mathrm{i}x\cdot\xi} f(x)\,\mathrm{d}x, \qquad D = -\mathrm{i}\nabla, \qquad U(T) = e^{-\mathrm{i}T|D|}.$$

Initially f is Schwartz. The generator $|D|$ is positive self-adjoint on $L^2(\mathbb{R}^n)$ with domain $H^1(\mathbb{R}^n)$. Its unitary group solves $(\partial_T + \mathrm{i}|D|)u = 0$ and consequently $(\partial_T^2 - \Delta)u = 0$, with the compatible initial velocity $\partial_T u(0) = -\mathrm{i}|D|f$. General real wave data require both half-wave branches, with their corresponding displacement and velocity norms. No spatial boundary condition is imposed in the Euclidean problem.

Put $d = n - 1$, $I = [1, 2]$, and

$$p_{\mathrm{c}} = \frac{2n}{n-1}, \qquad s_+(p) = \max\left\{0, \frac{n-1}{2} - \frac{n}{p}\right\}, \qquad J = (1-\Delta)^{1/2}. \tag{1}$$

The source's Theorem 1.1 asserts, for every $n \geq 2$, $2 \leq p < \infty$, and $\epsilon > 0$,

$$\|U(T)f\|_{L^p(\mathbb{R}^n \times I)} \leq C_{n,p,\epsilon} \left\| J^{s_+(p)+\epsilon} f \right\|_{L^p(\mathbb{R}^n)}. \tag{2}$$

At $p = 2$ no ϵ is needed, by unitarity. The stronger finite-scale claim, its Theorem 1.2, is

$$\|U(T)\varphi(D/N)f\|_{L^{p_c}(\mathbb{R}^n \times I)} \leq C[\log(2+N)]^{2/p_c} \|f\|_{p_c}, \qquad N \geq 2, \tag{3}$$

for a fixed smooth annular cutoff φ. The constant must be independent of N and of the input. An estimate for a fixed finite frequency bank does not establish this quantifier.

2.1 What local smoothing does and does not mean

The gain concerns an L^p norm after integration over time. It does not assert heat-like regularization of every rough input at each fixed time. At frequency N, the standard fixed-time loss for $p \geq 2$ is $(n-1)(1/2 - 1/p)$ derivatives. Above p_c, the expected spacetime order in (1) improves this by $1/p$. At p_c that order is zero, but the source's radial focusing example requires at least a logarithmic power $1/p_c$. Thus (3) leaves a gap between $1/p_c$ and $2/p_c$ for unrestricted data.

2.2 Prior and recent literature

Guth–Wang–Zhang established sharp cone square-function estimates and the local-smoothing conjecture in two spatial dimensions [3]. Bourgain–Demeter decoupling is a classical higher-dimensional input [4]. The revised January 2026 paper of Gan–He–Li–Wu gives, in particular, the Euclidean three-dimensional range $p \geq 10/3$, not the new manuscript's critical $p = 3$ logarithmic estimate [5]. The distinction between Euclidean and variable-coefficient or manifold thresholds is essential.

Two recent primary sources are especially relevant to possible extensions. Rozendaal–Schippa study rough-coefficient wave equations using bilinear and decoupling methods [6]. Their coefficient regularity assumptions cannot be dropped merely by switching to Ram coordinates. Cushman–Demeter–Wu investigate diameter-free reverse inequalities and superorthogonality, including a sharp logarithmic square-function loss for a canonical cone partition [7]. Such an estimate is not automatically the same as (3); the input norm, angular partition, and reassembly must be checked.

3 A dependency map of the source argument

Table 1 records the principal mathematical functions of the source sections. It avoids conflating an evolution formula, an auxiliary elliptic solve, and a norm inequality.

Table 1: Source dependencies and the legitimate Ram interface.

Source	Essential step	Ram interface and retained obligation
2	Symmetric system, imaginary translation, positive conserved current	Ordered transforms and coherent amplitude representation; retain polarization, conjugation, and column divergence.
3	Quaternionic Hessian, Moore cofactors, real divergence and Green identity	Real-block realization and weak testing; the Moore determinant is not an ordinary determinant of noncommuting entries.

Source	Essential step	Ram interface and retained obligation
4	Rank-one fiber PDE, Schur complement, moving-wall energy cancellation	Boundary-aware nonlinear residuals; retain the full spacetime Hessian and the null-line geometry of the wall.
5	Gaussian wall decay, inverse multiplier in the interior, quadratic comparison	Explicit window conditioning and residual budgets; all constants must be tracked as $h = N^{-1} \to 0$.
6	Two radial harmonic comparisons on a thin cylinder	Exact logarithmic-coordinate ODE; scalar supersolution sharpness prevents a generic removal of one logarithm.
7	Torus-to-Euclidean transfer, interpolation, dyadic summation	Physical reconstruction and tails; finite-frame isometries alone do not give a critical L^p estimate.
8	Radial focusing and weak-spacetime obstructions	Mandatory negative controls for any proposed stronger estimate.
9	Wave-to-sphere extension, random-sign caps, Kakeya and Nikodym	Retain arbitrary tube centers and direction weights; transformed coefficients do not remove incidence geometry.
10–12	Bochner–Riesz, Stein square function, paraboloid and Schrödinger consequences	Trace each strict exponent, summation, and epsilon-removal hypothesis; these remain downstream of the wave bound.
A	Periodic quaternionic Dirichlet solvability and parameter dependence	Full cofactor Jacobian with Dirichlet closure; qualitative existence is not wavelength-uniform conditioning.
B	Angularly regular endpoint estimates	Angular-weighted domains give a genuine restricted improvement already present in the source.
C	Disjoint-input packets becoming overlapping tubes	Coherent reconstruction must preserve phases and the weak-norm obstruction.
D	A separate linear disk PDE when $n = 2$	Cleanest test case for boundary-aware Ram panels; does not linearize all dimensions.
E	Simple extremal curved characteristic sheets of symmetric pencils	Ordered matrix symbols and spectral gaps; no automatic extension to arbitrary sheets or variable coefficients.

The bounded audit carried out here rederived the central algebraic cancellation, checked the comparison directions and the principal boundary-rotation calculation, and found no demonstrated fatal error in those checked steps. This statement is deliberately narrower than certification of the source theorem. The complete boundary-regularity localization, parameter-dependent functional analysis, and every frequency-uniform multiplier estimate were not independently proved in full. None of the numerical checks below fills those analytic obligations.

4 Ram operators: definitions that can safely be transferred

4.1 Kernel moments and finite remainders

The revised TR62 and RamOperator foundational report [14, 15] describe a kernel operator in displacement variables:

$$(Kf)(x) = \int h(x,\eta) f(x-\eta)\,\mathrm{d}\eta, \qquad S_\alpha(x) = \frac{(-1)^{|\alpha|}}{\alpha!} \int \eta^\alpha h(x,\eta)\,\mathrm{d}\eta. \tag{4}$$

For $f \in C^{M+1}$ on all relevant line segments and integrable absolute remainders, integrated Taylor expansion gives

$$Kf = \sum_{|\alpha|\le M} S_\alpha \partial^\alpha f + R_M, \quad |R_M(x)| \le \sum_{|\alpha|=M+1} \frac{1}{\alpha!} \int |h(x,\eta)\eta^\alpha| \sup_{0\le t\le 1} |\partial^\alpha f(x - t\eta)| \, \mathrm{d}\eta. \tag{5}$$

This is the Ram moment differential representation. For fixed coefficients it is linear. The nonlinear map $u \mapsto \det_{\mathrm{M}} H_{\mathrm{fib}}(B_0 + u)$ is not this operator.

When differentiation under the integral is justified and the additional derivatives of f and the kernel are integrably controlled, differentiating (5) uses every Leibniz term:

$$\partial^\beta (Kf) = \sum_{|\alpha|\le M} \sum_{\gamma \le \beta} \binom{\beta}{\gamma} (\partial^{\beta-\gamma} S_\alpha) \partial^{\alpha+\gamma} f + \partial^\beta R_M. \tag{6}$$

A finite jet inverse exists only after specifying a trial space, closing the higher derivatives, imposing boundary or compatibility rows, and proving the resulting matrix is nonsingular with a useful stability margin. Normalizing a kernel does not imply that matrix is invertible.

4.2 Nonanalytic windows are not infinite Taylor jets

An alternative in the corrected RamOperator series is

$$R_k^{(b)} f(x) = \frac{2k+1}{2b} \int_{-b}^{b} f(x+a) P_k(a/b) \, \mathrm{d}a, \quad f(x+a) = \sum_{k\ge 0} R_k^{(b)} f(x) P_k(a/b) \quad \text{in } L^2([-b,b]). \tag{7}$$

The integral representation needs window L^2 data, not analyticity. Its derivative-series realization requires extra convergence hypotheses. This distinction matters because the source uses smooth compact cutoffs, which are generally not analytic. A local point jet of an arbitrary C^∞ function does not determine all its neighboring values.

For a constant scalar moment operator on polynomials of degree at most N, $K = m_0 I + B$ with degree-lowering B has

$$K^{-1} = m_0^{-1} \sum_{j=0}^{N} (-B/m_0)^j, \qquad m_0 \neq 0.$$

This finite algebraic identity is valuable but is not a bounded inverse on an infinite-dimensional function space. In particular a second-order elliptic expression annihilates constants before boundary conditions are imposed. Its unrestricted polynomial matrix has a zero constant column.

4.3 Which additional corpus results matter

TR216 supplies differentiated test atoms and boundary terms; TR248 supplies finite-frame leakage and perturbation margins; TR250 supplies compatible weak forms and physical metric bookkeeping [20, 21, 22]. TR205's determinant analysis and TR-item-293's branch examples warn that small determinant residuals need not select the admissible elliptic branch [23, 24]. The four new RMT-extension reports add:

1. support-preserving frame normalization and a specified L^p patch coefficient space [16];
2. graph-domain and spectral-power discipline, with compression and commutator effects [17];

3. ordered hypercomplex channels and constrained observability [18];

4. full nonlinear encoders, decoders, and resolved–tail validation [19].

These are substantive tools. None is an unconditional substitute for the critical geometric estimate in (3).

5 Null coordinates and an exact heat–Schrödinger reformulation

Write the physical spatial coordinate as $X = (x, b)$, $x \in \mathbb{R}^d$, and put

$$t = (T + b)/\sqrt{2}, \qquad s_0 = (b - T)/\sqrt{2}.$$

Then $\Delta_X - \partial_T^2 = 2\partial_t\partial_{s_0} + \Delta_x$. For a plane wave away from the backward ray, set

$$\kappa = (\xi_b + |\xi|)/\sqrt{2} > 0, \qquad v = \xi_x/\kappa.$$

A direct substitution gives

$$u_{\kappa,v} = c_{\kappa,v} e^{\mathrm{i}\kappa(s_0 + v\cdot x - |v|^2 t/2)}. \tag{8}$$

At fixed κ, the coefficient of $e^{\mathrm{i}\kappa s_0}$ solves $(2\mathrm{i}\kappa\partial_t + \Delta_x)u_\kappa = 0$. This is an exact null-frequency decomposition, not a paraxial approximation.

For $W \in \mathbb{R}^d$ and $y > 0$ define the coherent Gaussian window

$$F_{\kappa,y,W} = \sum_v c_{\kappa,v} e^{\mathrm{i}\kappa(s_0 + v\cdot x - |v|^2 t/2)} e^{-\kappa y|v-W|^2/2}. \tag{9}$$

For finite sums, all differentiations are literal. For general band-limited L^2 inputs, spectral calculus justifies the same statements.

Proposition 5.1 (Exact covariant heat window)**.** *At fixed κ, W,*

$$F_{\kappa,y,W} = \exp\left\{\frac{y}{2\kappa}(\nabla_x - \mathrm{i}\kappa W)^2\right\} u_\kappa, \tag{10}$$

$$\partial_y F_{\kappa,y,W} = \frac{1}{2\kappa}(\nabla_x - \mathrm{i}\kappa W)^2 F_{\kappa,y,W}, \tag{11}$$

$$\partial_{W_j} F_{\kappa,y,W} = y(D_{x_j} - \kappa W_j) F_{\kappa,y,W}. \tag{12}$$

The window commutes with the fixed-κ Schrödinger evolution.

Proof. On the mode (8), $\nabla_x - \mathrm{i}\kappa W$ acts by $\mathrm{i}\kappa(v - W)$. Its squared symbol is $-\kappa^2|v - W|^2$, giving the exponential and the y derivative. Differentiating the Gaussian in W_j gives $\kappa y(v_j - W_j)$, equal to the last displayed operator. All multipliers commute at fixed parameters. □

The source's normalized multiplier also contains $(\kappa/y)^{d/2}$. Consequently the normalized field obeys an additional $-d/(2y)$ zeroth-order term in its y equation. Omitting it would be an incorrect normalization. If $W = A + \mathbf{k}B$ in the source's auxiliary complex plane, $|W - v|^2 = |A - v|^2 + |B|^2$ supplies a further real attenuation. Different κ components have different Schrödinger coefficients; they must not be replaced by one universal fixed-κ equation.

Equations (9)–(12) expose a useful Ram window/moment interface: parameter derivatives encode centered spectral moments, and propagation preserves their exact phase relation. They rederive the window through classical semigroup calculus. They do not solve the separate nonlinear auxiliary PDE.

6 Imaginary continuation as a contractive spectral filter

The paper evaluates finite waves at imaginary light-ray shifts. For a unit vector ω and $\rho \geq 0$,

$$e^{\mathrm{i}(X\cdot\xi - T|\xi|)}\big|_{(T,X)\mapsto(T,X)-\mathrm{i}\rho(1,\omega)} = e^{\mathrm{i}(X\cdot\xi - T|\xi|)} e^{-\rho|\xi||\widehat{\xi}-\omega|^2/2}. \tag{13}$$

The sign is important: it gives attenuation, not growth.

Define $A_\omega = |D| - \omega \cdot D$ by its nonnegative real Fourier symbol. Its maximal self-adjoint domain is $\{f \in L^2 : (|\xi| - \omega\cdot\xi)\widehat{f} \in L^2\}$, which contains H^1 and need not equal it. Put $M_\rho = e^{-\rho A_\omega}$.

Theorem 6.1 (Bandwise moment and inverse certificate)**.** *M_ρ is an L^2 contraction semigroup. On*

$$E = \{\xi : |\xi| \leq C_0 N,\ |\widehat{\xi} - \omega| \leq \theta\}, \qquad a_* = \tfrac{1}{2} C_0 N \theta^2, \quad \delta = \rho a_*,$$

its inverse on $L^2(E)$ has norm at most e^δ. Moreover

$$\left\| M_\rho f - \sum_{j=0}^{m} \frac{(-\rho A_\omega)^j}{j!} f \right\|_2 \leq \frac{\delta^{m+1}}{(m+1)!} \|f\|_2, \qquad \operatorname{supp} \widehat{f} \subset E. \tag{14}$$

If data error is at most η and a multiplier approximation has operator error at most $\epsilon < e^{-\delta}$ on E, inversion with that approximate multiplier has error at most

$$\frac{\eta + \epsilon \|f\|_2}{e^{-\delta} - \epsilon}. \tag{15}$$

Proof. The symbol $a(\xi) = |\xi|(1 - \widehat{\xi}\cdot\omega)$ is nonnegative, and $|\widehat{\xi} - \omega|^2 = 2(1 - \widehat{\xi}\cdot\omega)$. On E, $0 \leq a \leq a_*$. The spectral theorem proves the contraction, semigroup law, and inverse bound. Taylor's integral remainder for e^{-x} on $0 \leq x \leq \delta$ is bounded by $x^{m+1}/(m+1)!$, since the absolute derivative e^{-t} is at most one. Plancherel gives (14). The perturbed multiplier is bounded below by $e^{-\delta} - \epsilon$. Applying its inverse to the data error plus the model error proves (15). □

For example, $\delta = 1$, $m = 6$ gives the rigorous error bound $1/7! = 1.98413 \times 10^{-4}$ and inverse amplification at most e. This useful bounded-band estimate does not authorize backward heat inversion on all frequencies. If $\rho N \theta^2$ grows, the inverse becomes exponentially ill-conditioned. Nor does an L^2 multiplier bound alone control the critical L^{p_c} multiplier norm: the source separately bounds derivatives of reciprocal symbols through a dimension-dependent order.

7 The coherent current and quaternionic Hessian

For the null coordinates, take real symmetric matrices

$$C^t = 2E_{00}, \quad C^{x_j} = E_{0j} + E_{j0}, \quad C^{s_0} = -\sum_{j=1}^{d} E_{jj}.$$

Writing $p = \kappa(-|v|^2/2, v, 1)$ gives

$$-C(p) = \kappa \begin{pmatrix} |v|^2 & -v^{\mathsf{t}} \\ -v & I_d \end{pmatrix}, \qquad -w^{\mathsf{t}} C(p) w = \kappa |w' - v w_0|^2. \tag{16}$$

Its null vector is $E(v) = (1, v)^{\mathsf{t}}$. The vector wave obtained by summing $\kappa^{d/2} E(v) u_{\kappa,v}$ solves the first-order system $\mathcal{P}F = \sum_\mu C^\mu \partial_\mu F = 0$. Equation (16) is precisely why continuation in the quadratic flux direction produces a Gaussian centered at the polarization.

The source builds two real state copies, evaluates the *conjugate* continued vector wave, and forms a positive tensor from the real parts of conjugate products of vector fields. Its divergence calculation uses $\mathcal{P}F = 0$ and a purely imaginary acceleration. Under the nonlinear holomorphic flux coordinate map, the tensor density transforms with the inverse Jacobian and two tangent factors. A second-derivative term in that transformation vanishes because the state blocks are paired and the coordinate map is holomorphic. Conservation is therefore not an automatic property of an arbitrary tensor coordinate change.

In the resulting quaternionic notation the current is

$$A^0 = \chi\chi^*, \qquad \chi = \sum_{\kappa,v} V^v \, \iota_e(\overline{u_{\kappa,v}}) \, (\kappa/y)^{d/2} e^{-\kappa y |W-v|^2/2}, \tag{17}$$

where V^v is the source's characteristic polarization in the flux chart, $e = Y/|Y|$, and ι_e is its complex-to-quaternion embedding. Its real columns are divergence-free and it is tangent to the passive coordinate Φ. The source's time density is the sum of squares of two scalar multiplier outputs, not the sum of the individual mode energies.

Example 7.1 (Why coherence cannot be discarded)**.** At a point where two equal weighted amplitudes are 1, their coherent density is $|1 + 1|^2 = 4$, while the modewise density is 2. For amplitudes $1, -1$, the coherent density is zero while the same modewise density remains 2. No positive diagonal reweighting of the individual energies reproduces both cases. Frame or channel compression must preserve cross terms, or explicitly bound their loss.

For a real potential v and three real slots per index, define

$$G_a(v) = \sum_{\alpha=0}^{2} e_\alpha \partial_{a\alpha} v, \qquad H_{ab}(v) = \sum_{\alpha,\beta=0}^{2} e_\alpha \overline{e_\beta} \, \partial_{a\alpha}\partial_{b\beta} v, \quad (e_0, e_1, e_2) = (1, \mathbf{i}, \mathbf{j}). \tag{18}$$

This is a quaternionic Hermitian matrix and $\operatorname{tr} H(v) = \Delta v$. For functions of the imaginary complex coordinates alone, $H = 4\bar\partial\partial$ in the source's ordered convention. Thus $H(|P|^2) = 4I$, whereas for the square of distance in all three slots per index $H(\ell^2) = 6I$. Confusing these two normalizations changes comparison constants.

The Moore determinant $\det_{\mathrm{M}} Q$ of a quaternionic Hermitian matrix is real. For $Q > 0$,

$$\mathcal{C}(Q) = \det_{\mathrm{M}}(Q) Q^{-1}, \quad D(\det_{\mathrm{M}})(Q)[E] = \operatorname{Re}\operatorname{tr}(\mathcal{C}(Q)E), \quad D(\log\det_{\mathrm{M}})(Q)[E] = \operatorname{Re}\operatorname{tr}(Q^{-1}E). \tag{19}$$

The order and real trace matter. The cofactor of a Hessian has a divergence-free real representation; the source gives a polarized constant-coefficient proof compatible with its three-slot setting. Related quaternionic cofactor divergence is part of the classical literature [9]. An unconstrained quaternionic matrix is not entitled to this determinant or divergence identity.

8 The auxiliary Dirichlet problem and its proof function

At wavelength h, put $a = Kh$, $\lambda = h^{-30n}$, and let the three real parameters be $z = (T, \tau)$, $\tau \in \mathbb{C}_{\mathbf{k}}$, $s = |\tau|$. On each spatial fiber use periodic real coordinates $X \in \mathbb{R}^n/(100\mathbb{Z})^n$ and bounded imaginary

coordinates (P, Z). Set

$$Y = (\tau + Z)/\sqrt{2}, \qquad W = P/Y, \tag{20}$$

$$\Phi = (Z - \tau)/\sqrt{2} + (P \cdot P)/(2Y), \qquad Q = \Phi/a, \tag{21}$$

$$r = |W|^2/w_0^2 + 100|Q|^2 - 1, \qquad w_0 = 1/10. \tag{22}$$

The product $P \cdot P$ is complex bilinear. The fiber Ω_z is $r < 0$ with the stated periodic real coordinates. Its inverse chart has

$$Y = \frac{\sqrt{2}\tau + aQ}{1 + W \cdot W/2}, \quad P = WY, \quad Z = \sqrt{2}Y - \tau.$$

For $a < s < 4$, $|Y| \asymp s$ and the imaginary volume element is comparable to $a^2 s^{2d}\, \mathrm{d}W\, \mathrm{d}Q$. The spatial Hessian satisfies $H_{\mathrm{fib}} r \gtrsim s^{-2} I$. The full Hessian $H(r)$ has a null line that is also tangent to the wall and annihilates the background.

With the source's passive cutoff $g(Q)$ and positive spatial smoothing kernel J_h, define

$$A_{\mathrm{w}} = g(Q)(J_h *_X A^0), \quad A = A_{\mathrm{w}} + h^{50n}\mathcal{C}(H(r)), \quad \mu = A_{TT} > 0, \quad B_0 = \lambda|\Phi|^2. \tag{23}$$

The core auxiliary equation is

$$\boxed{D_u = H_{\mathrm{fib}}(B_0 + u) > 0, \qquad \det_{\mathrm{M}} D_u = \lambda\mu \text{ in } \Omega_z, \qquad u|_{\partial\Omega_z} = 0.} \tag{24}$$

The unknown u is real, with periodicity in X. There is no free constant gauge because the lateral Dirichlet condition fixes it. The background $B_{\mathrm{fib}} = H_{\mathrm{fib}} B_0$ has rank one, not full rank; its positive eigenvalue is comparable to λ.

An identity-background formula is possible only with changed boundary data: if $q = (|P|^2 + |Z|^2)/4$ and $v = B_0 + u - q$, then $\det_{\mathrm{M}}(I + H_{\mathrm{fib}} v) = \lambda\mu$, but $v|_{\partial\Omega_z} = B_0 - q$, generally not zero. Silently substituting a homogeneous boundary condition for v would change the problem and its energy geometry.

8.1 Existence assumptions and the role of Appendix A

The source's Appendix A assumes a smooth bounded imaginary domain, a defining function with strictly positive quaternionic Hessian on the closure, a nonnegative background Hessian, and a smooth prescribed density strictly larger than the background determinant. It asserts smooth admissible solvability, uniqueness, comparison barriers, and smooth dependence on the fiber parameters.

Comparison linearizes the difference of two log determinants through a positive segment. The continuity argument starts at a known strictly admissible defining potential. Boundary mixed derivatives are controlled using local quaternionic rotations after adding a periodic fourth slot. If ν is the quaternionic unit real normal and v a real tangent vector, the skew-Hermitian matrix

$$S = v\nu^* - \nu v^* - \nu(\nu^* v)\nu^*$$

satisfies $S\nu = v$. The last term is needed because real tangency gives $\mathrm{Re}(\nu^* v) = 0$, not necessarily $\nu^* v = 0$. The covariance commutator cancels after pairing with the inverse Hessian. A Schur complement then isolates the remaining normal second derivative.

These checks support the algebra of that boundary argument. Classical quaternionic Dirichlet results [8, 10] are relevant, but a bounded Euclidean-domain theorem alone is not a literal theorem

on the periodic quotient with moving fibers. The source's adaptation and regularity argument must still be used. The cutoff applied to the nonlinear operator in Appendix A is identically one near the controlled solution-Hessian range; loss of global ellipticity outside that range alone does not refute the cited range-dependent regularity theorem. No counterexample to that step was established in this investigation. Most importantly, these qualitative existence constants need not be uniform in h.

9 Rederivation of the determinant-energy cancellation

The full Hessian, including the time parameter slots, has block form

$$H(B_0+u)=\begin{pmatrix}\ell & c^*\\ c & D_u\end{pmatrix}=\sigma_u e_T e_T^*+\Sigma_u,\qquad \sigma_u=\ell-c^*D_u^{-1}c,\qquad \Sigma_u=\begin{pmatrix}c^*D_u^{-1}c & c^*\\ c & D_u\end{pmatrix}\geq 0.$$

No positivity of the full Hessian is claimed. Since $A:H(B_0)=0$, $A_{TT}=\mu$, and $\det_{\mathrm{M}} D_u=\lambda\mu$,

$$A:H(u)=A:\Sigma_u+\mu\sigma_u=A:\Sigma_u+\lambda^{-1}\det_{\mathrm{M}} H(B_0+u). \tag{25}$$

Here $A:B=\operatorname{Re}\operatorname{tr}(AB)$.

Define, on a fixed fiber,

$$X_0(z)=\int_{\Omega_z}(-u)\mu\,\mathrm{d}\zeta, \tag{26}$$

$$E_z(v)=\lambda^{-1}\int_{\Omega_z} v\int_0^1 \det_{\mathrm{M}} H_{\mathrm{fib}}(B_0+\theta v)\,\mathrm{d}\theta\,\mathrm{d}\zeta, \tag{27}$$

$$\mathcal{J}_z(v)=E_z(v)-\int_{\Omega_z} v\mu\,\mathrm{d}\zeta,\qquad \mathcal{F}(z)=\mathcal{J}_z(u). \tag{28}$$

The segment determinant is a polynomial, so the singular rank-one endpoint $\theta=0$ causes no division by zero.

9.1 Why the moving boundary cannot be suppressed

For a compactly supported parameter test $\eta(z)$, the only nonzero entry of $H(\eta)$ is $H_{TT}(\eta)=\Delta_z\eta$. Green's identity on the *total* fibered domain gives, for a divergence-free real tensor representation M,

$$\int_\Omega uM_{TT}\Delta_z\eta=\int_\Omega \eta\,M:H(u)-\int_{\partial_{\mathrm{lat}}\Omega}\eta\,(\nu^*M\nu)\partial_{\mathbf{n}}u. \tag{29}$$

The zero trace removes the term involving $u\nabla\eta$; it does not remove every normal flux. Writing $u=r\rho$ gives $\partial_{\mathbf{n}}u=\rho|\nabla r|$ on the wall.

Apply (29) first with $M=A$ and the negative sign in X_0. It yields a wall flux minus $\int A:\Sigma_u$ minus the full determinant in (25). Apply it next with $M=\mathcal{C}(H(B_0+\theta u))$ and integrate over θ. On the wall, the compressed Hessian retains the common null line of $H(r)$ and $H(B_0)$ inside $\nu^\perp$. The cofactor normal entry therefore vanishes. The second application gives the opposite full determinant. Thus

$$\Delta_z\mathcal{F}+\int_{\Omega_z}A:\Sigma_u\,\mathrm{d}\zeta=\int\rho\,G(r)^*A_{\mathrm{w}}G(r)\delta(r)\,\mathrm{d}\zeta. \tag{30}$$

The gradient is the full gradient, and $\delta(r)\,\mathrm{d}\zeta=\mathrm{d}S_{\mathrm{fib}}/|\nabla_{\mathrm{fib}}r|$. The regularizing cofactor has no wall flux. This is a structural cancellation, not generic integration by parts on a fixed domain.

9.2 Energy positivity

For zero-trace variations w and admissible D_v,

$$\mathcal{J}_z'(v)[w] = \int w\{\lambda^{-1} \det_{\mathrm{M}} D_v - \mu\}\, \mathrm{d}\zeta, \tag{31}$$

$$\mathcal{J}_z''(v)[w,w] = -\lambda^{-1} \int G_{\mathrm{fib}}(w)^* \mathcal{C}(D_v) G_{\mathrm{fib}}(w)\, \mathrm{d}\zeta. \tag{32}$$

The cofactor divergence identity and zero boundary values justify interchanging the two test potentials when differentiating (28). Hence the exact admissible solution is the unique maximizer in the convex admissible class.

Rank one gives the exact polynomial identity

$$\det_{\mathrm{M}} D_{\theta u} = \theta^n \det_{\mathrm{M}} D_u + (1-\theta)\theta^{n-1} \mathcal{C}(D_u) : B_{\mathrm{fib}}.$$

Integrating it and using the Euler identity $\mathcal{C}(D_u) : D_u = n \det_{\mathrm{M}} D_u$ proves

$$\mathcal{F} = \frac{n-1}{n+1} X_0 + \frac{1}{\lambda n(n+1)} \int G_{\mathrm{fib}}(u)^* \mathcal{C}(D_u) G_{\mathrm{fib}}(u)\, \mathrm{d}\zeta, \qquad \frac{n-1}{n+1} X_0 \le \mathcal{F} \le \frac{n}{n+1} X_0. \tag{33}$$

These identities are independently reproducible conditional on the smooth admissible solution, current conservation, and the stated fiber geometry. They are not an independent proof of all of those hypotheses.

10 TR60 and TR3: a substantive quaternionic Ram connection

10.1 The relevant reports and corrected scope

The dedicated original report is TR60, *Quaternionic RMT*, in `TR60-Quaternionic-RMT/Quaternionic_RMT.tex`. For operative claims this investigation uses its September 19, 2026 foundation revision, `TR-rev-1-old-tr060/TR-rev-1-old-tr060.tex`, alongside the October Volume 12 treatment and the October 23 *TR3-rmt-ext-hypercomplex-constraints* report [12, 13, 25, 18]. These sources do provide a genuine quaternionic extension of Ram Calculus. The distinction needed here is between what that extension proves and what the nonlinear fiber equation additionally requires.

In the corrected TR60 convention, for real center d_0, positive scales, a real window, and integrable product of factor norms, a historically named left-linear transform is

$$F_f^L(p) = \sigma^{-N/2} \int_{\mathbb{R}^N} w(z; d_0, W_0)\, f(z)\, \psi_q((z-d_0)/\sigma)\, E_q^r(A_L(z))\, \mathrm{d}z. \tag{34}$$

The full kernel is *on the right* of f, which makes this transform left quaternion-linear in its signal. The reversed convention reverses the complete integrand order, not only one exponential. Quaternionic Mittag–Leffler coefficients have specified sides:

$$E_q^r(\zeta) = \sum_{k\ge 0} \zeta^k \Gamma_{\mathbb{H}}(kq+1)^{-1}, \qquad E_q^\ell(\zeta) = \sum_{k\ge 0} \Gamma_{\mathbb{H}}(kq+1)^{-1} \zeta^k.$$

Branches and terminals are required for noninteger orders. The relevant wave and Hessian calculations here use ordinary integer real derivatives, so no unproved fractional translation expansion is needed.

The corrected report restricts its unconditional Fourier Plancherel specialization to a fixed quaternion axis and $q = 1$. In one such right-kernel convention,

$$\mathcal{F}_{\mathbf{n}} f(\xi) = \int f(x) e^{-\mathbf{n}\, x\cdot\xi}\, \mathrm{d}x, \quad \mathbf{n}^2 = -1, \qquad \|\mathcal{F}_{\mathbf{n}} f\|_2^2 = (2\pi)^N \|f\|_2^2 . \tag{35}$$

If $f = f_1 + f_2\mathbf{m}$ with $f_j \in \mathbb{C}_{\mathbf{n}}$ and $\mathbf{mn} = -\mathbf{nm}$, the second complex component has the opposite Fourier sign. For a general quaternionic kernel both components couple. Arbitrary quaternion-order atoms do not inherit (35) or a continuum inverse merely from their formal series.

10.2 Real blocks, metrics, and full operator order

Let $\mathrm{vec}(q) = (q_0, q_1, q_2, q_3)^{\mathsf{t}}$. Define $L(p)$ and $R(q)$ by

$$\mathrm{vec}(pxq) = L(p)R(q)\,\mathrm{vec}(x).$$

Quaternion associativity gives

$$L(pq) = L(p)L(q), \quad R(pq) = R(q)R(p), \quad L(p)R(q) = R(q)L(p), \tag{36}$$

$$L(q)^{\mathsf{t}} = L(\bar{q}), \quad R(q)^{\mathsf{t}} = R(\bar{q}), \quad L(q)^{\mathsf{t}}L(q) = |q|^2 I. \tag{37}$$

Two left multiplication blocks need not commute. The two-sided map $x \mapsto p(xq)$ is generally real-linear, not right $\mathbb{H}$-linear: its defect on multiplication by r is $p[x(rq - qr)]$. For nonzero p it is right-linear for every r only when q is real.

For physical input and output metrics W, V, the real adjoint is $B^\sharp = W^{-1}B^{\mathsf{t}}V$. A quaternionic Hermitian $n \times n$ matrix has $2n^2 - n$ real degrees of freedom, with

$$\|A\|_F^2 = \mathrm{Re}\,\mathrm{tr}(A^2) = \sum_a A_{aa}^2 + 2\sum_{a<b} |A_{ab}|^2. \tag{38}$$

Each real component of an off-diagonal quaternion therefore carries weight two in a packed-coordinate representation. Treating all packed coordinates as equally weighted changes the physical norm.

These blocks exactly realize the paper's cofactor quadratic form:

$$C^{\mathbb{R}}_{a\alpha,b\beta} = \mathrm{Re}(\bar{e}_\alpha C_{ab} e_\beta), \qquad C : H(w) = \mathrm{div}(C^{\mathbb{R}}\nabla w) \quad \text{when } C = \mathcal{C}(H(v)). \tag{39}$$

For three slots, take the compression to $(1, \mathbf{i}, \mathbf{j})$ at each index. This is an exact connection between the corrected Ram real-block calculus and the PDE, not a scalar-symbol approximation.

10.3 TR3's constrained observation theorem and its boundary

In whitened real coordinates, let P project onto an admissible closed *linear* signal space, B_s be ordered channel maps, P_s output projections, and $F_{s,N}$ scalar analysis maps with $F_{s,N}^* F_{s,N} = I - \mathcal{L}_s$, $0 \le \mathcal{L}_s \le I$. The actual observation map is $\mathcal{A}_N f = (F_{s,N} P_s B_s f)_s$, $f \in \mathrm{ran}\, P$. TR3 gives the exact identity

$$\mathcal{A}_N^* \mathcal{A}_N = P(G - D_{\mathrm{loss}} - E_{\mathrm{loss}})P\big|_{\mathrm{ran}\, P}, \tag{40}$$

$$G = \sum_s B_s^* B_s, \qquad D_{\mathrm{loss}} = \sum_s B_s^*(I - P_s)B_s, \tag{41}$$

$$E_{\mathrm{loss}} = \sum_s B_s^* P_s \mathcal{L}_s P_s B_s. \tag{42}$$

For an isometric synthesis V of a specified finite physical trial space, a sufficient lower observation margin is

$$\gamma^2 = \lambda_{\min}(V^*GV) - \|V^*D_{\mathrm{loss}}V\| - \|V^*E_{\mathrm{loss}}V\| > 0.$$

Then the reconstruction pseudoinverse has norm at most $1/\gamma$. This separates the loss caused by deleting channel components from the loss caused by retaining finitely many scalar coefficients.

Hermitian symmetry and a fixed tangency constraint can fit this linear framework. Positivity $Q > 0$, the equation $\det_{\mathrm{M}} Q = \lambda\mu$, and the assertion that $Q - B_{\mathrm{fib}}$ is the Hessian of one scalar potential do not follow from it. Spectral clipping of a reconstructed matrix field can destroy Hessian compatibility and cofactor divergence. A differential divergence constraint also requires a closed operator domain and its boundary conditions; its projection need not preserve local support.

Example 10.1 (Complementary quaternion channels)**.** On pure imaginary quaternions, the projected map $P_{\mathrm{im}}L(\mathbf{n})P_{\mathrm{im}}$ is $v \mapsto \mathbf{n} \times v$. For axes $\mathbf{i}, \mathbf{j}$ their combined Gram matrix is $\mathrm{diag}(1, 1, 2)$. Each channel is singular, but the pair has lower gain one before truncation. This TR3 example is useful for channel design, not a proof that a sampled Hessian is positive or that a wave-current reconstruction is divergence-free.

10.4 A new explicit order-error bound for this current

The embedding in (17) is $\iota_e(\alpha + \mathrm{i}\beta) = \alpha + e\mathbf{i}\beta$, where $e \in \mathbb{C}_{\mathbf{k}}$ has unit norm. Its polarization factors belong to $\mathbb{C}_{\mathbf{k}}$, a generally different slice.

Proposition 10.2 (Mixed-slice order defect and current error)**.** *For real a, b, α, β,*

$$[a + \mathbf{k}b, \alpha + e\mathbf{i}\beta] = 2b\beta\, e\mathbf{j}, \qquad |[a + \mathbf{k}b, \alpha + e\mathbf{i}\beta]| = 2|b\beta|. \tag{43}$$

If $\widetilde{\chi}$ is obtained by reversing those factors mode by mode in (17), then

$$\|\chi - \widetilde{\chi}\| \le 2\sum_{\kappa,v} m_{\kappa,v}|\beta_{\kappa,v}|\, \|\mathrm{Im}_{\mathbf{k}}\, V^v\|\,, \tag{44}$$

$$\|\chi\chi^* - \widetilde{\chi}\widetilde{\chi}^*\|_F \le (\|\chi\| + \|\widetilde{\chi}\|)\, \|\chi - \widetilde{\chi}\|\,. \tag{45}$$

All norms in the first line are quaternion vector norms, and the Gaussian $m_{\kappa,v}$ is real.

Proof. The $\mathbb{C}_{\mathbf{k}}$ factors commute with e and satisfy $\mathbf{i}p = \bar{p}\,\mathbf{i}$. Thus $p(e\mathbf{i}) - (e\mathbf{i})p = e(p - \bar{p})\mathbf{i} = 2be\mathbf{j}$. Summation and the triangle inequality give the first bound. For the second add and subtract $\widetilde{\chi}\chi^*$ and use $\|ab^*\|_F = \|a\|\,\|b\|$. □

This identifies a concrete implementation error that a scalar treatment can miss. It does not imply that every order change is observable in current data. For a common unit right phase r, $(\chi r)(\chi r)^* = \chi\chi^*$ exactly. The derivative $\delta\chi \mapsto \delta\chi\chi^* + \chi\delta\chi^*$ annihilates $\delta\chi = \chi s$ for $\mathrm{Re}\, s = 0$. TR3's linear observability certificate must therefore be applied to the appropriate linearization or lift, with this gauge accounted for, not blindly to the nonlinear Gram observation.

11 Boundary-aware Ram reformulation of the nonlinear solve

Let $X = C^{2,\alpha}_{D,\mathrm{per}}(\overline{\Omega_z}; \mathbb{R})$ denote real potentials with zero lateral trace and the prescribed periodicity. On the open admissible subset where $D_v > 0$, set

$$\mathcal{R}(v) = \det_{\mathrm{M}} D_v - \lambda\mu, \qquad \mathcal{L}(v) = \log \det_{\mathrm{M}} D_v - \log(\lambda\mu).$$

The density and geometry are held fixed in this fiber solve. Then

$$D\mathcal{R}(v)[w] = \mathcal{C}(D_v) : H_{\mathrm{fib}} w, \tag{46}$$

$$D\mathcal{L}(v)[w] = D_v^{-1} : H_{\mathrm{fib}} w. \tag{47}$$

In particular there is no extra zeroth-order $+w$ term: such a term in a different transport Monge–Ampère equation cannot be imported here. The determinant Newton correction obeys $\mathcal{C}(D_v) : H_{\mathrm{fib}} w = -\mathcal{R}(v)$ with homogeneous Dirichlet trace. For the logarithmic equation the right side is $-\mathcal{L}(v)$.

Proposition 11.1 (Coercive conforming Ram-panel Newton matrix)**.** *Suppose $cI \preceq D_v \preceq CI$ on the closed fiber. Let $\phi_1, \dots, \phi_m$ be linearly independent real conforming trial functions, periodic in X and zero on the lateral wall. Define*

$$K_{ij} = \int_{\Omega_z} \mathrm{Re}\{G_{\mathrm{fib}}(\phi_i)^* \mathcal{C}(D_v) G_{\mathrm{fib}}(\phi_j)\}\, \mathrm{d}\zeta, \quad K_{\Delta,ij} = \int \nabla\phi_i \cdot \nabla\phi_j \, \mathrm{d}\zeta.$$

Then

$$c^{n-1} K_\Delta \preceq K \preceq C^{n-1} K_\Delta. \tag{48}$$

For the determinant Newton step the correct load is $\int \phi_i \mathcal{R}(v)\, \mathrm{d}\zeta$.

Proof. The eigenvalues of $\mathcal{C}(D_v)$ are products of $n-1$ positive eigenvalues of D_v, hence lie between c^{n-1} and C^{n-1}. Realification and compression preserve the quadratic-form inequalities. Also $\|G_{\mathrm{fib}} w\|^2 = |\nabla w|^2$ for real w. Integration gives (48). Poincaré and independent zero-trace trial functions make K_Δ positive definite. Integrating the Newton differential equation by parts reverses its minus sign, giving the stated load. □

Polynomial or windowed Ram coordinates can assemble this same weak operator. They do not alter its physical coercivity. A sufficient positivity-preserving step rule for $v_{\mathrm{new}} = v + tw$ is $t\, \|H_{\mathrm{fib}} w\|_{L^\infty,\mathrm{op}} < c$. This is not a global Newton convergence theorem. Numerical quadrature must preserve or certify the matrix bounds, and a small finite set of residual tests must not be mistaken for the full residual norm.

11.1 What TR4 adds to a valid nonlinear implementation

TR4 [19] requires the complete physical residual after decoding, including off-range coefficients and unresolved tails. For a fixed injective encoder E with decoder D, $DE = I$, and $P = ED$, the model equation may be lifted as

$$E\mathcal{L}(Dc) + (I - P)c = 0,$$

provided the encoder on the residual space is also an injective specified realization. Its zeros then have no off-range component and solve the physical equation. Componentwise determinants of arbitrary encoded entries are a different nonlinear map.

If atoms or their normalizations depend on the current potential, the derivative contains both $E(v)D\mathcal{L}(v)[w]$ and $DE(v)[w]\mathcal{L}(v)$. The latter disappears only at a true root. Moving-fiber coordinates also differentiate Jacobians and boundary traces. These facts mirror the moving-domain terms in (29).

For a preconditioned Banach-space residual $F : X \to X$, suppose an approximate tangent $\widehat{J}$ is a genuine bounded isomorphism on the full resolved–tail space, with inverse norm at most β. If

$$\|F(v)\| \le \eta, \quad \left\| DF(v) - \widehat{J} \right\| \le \epsilon_1, \quad \|DF(x) - DF(y)\| \le L\, \|x - y\|,$$

throughout an admissible radius-r ball, then the sufficient conditions

$$\beta\eta + (\beta\epsilon_1 - 1)r + \tfrac{1}{2}\beta L r^2 \le 0, \qquad \beta\epsilon_1 + \beta L r < 1 \tag{49}$$

give a unique zero there by contraction of $x - \widehat{J}^{-1}F(x)$. The proof is the integral Taylor bound and Banach's theorem. The ball must remain in the positive cone. A finite-rank approximate inverse alone is not injective on the infinite-dimensional residual space: a fixed point would only prove that the *tested* residual vanishes. TR4's ambient tail and Schur conditions are useful precisely because they identify this missing obligation.

For an exact warning, let S be the right shift on ℓ^2, take $T = S^*$, and set $F(x) = Sx + e_1$. Then $x - TF(x) = 0$ for every x, so the iteration is a perfect contraction, but $F(0) = e_1 \neq 0$. The failure is the nontrivial kernel of the residual preconditioner, not a failure of Banach's fixed-point theorem.

12 Proved residual certificates on an admissible fiber

Let $V = H^1_{D,\mathrm{per}}(\Omega_z;\mathbb{R})$ with norm $\|w\|_V = \|\nabla w\|_2$, and let V^* carry its dual norm. The nonempty lateral boundary gives a Poincaré constant C_P. The following results do not depend on choosing Ram rather than conventional conforming coordinates.

Theorem 12.1 (Residual-to-solution bound)**.** *Let u solve (24), and let v be a smooth admissible potential with the same boundary and periodicity. If $D_u, D_v \succeq cI$, then*

$$\|v-u\|_V \le c^{-(n-1)}\,\|\mathcal{R}(v)\|_{V^*}, \qquad \|v-u\|_2 \le C_P c^{-(n-1)}\,\|\mathcal{R}(v)\|_{V^*}. \tag{50}$$

Proof. Put $e = v - u$ and $\overline{C} = \int_0^1 \mathcal{C}(D_{u+te})\,\mathrm{d}t$. The segment is positive and $\overline{C} \succeq c^{n-1}I$. The fundamental theorem of calculus and cofactor divergence give $\mathcal{R}(v) = \mathrm{div}(\overline{C}^{\mathbb{R}}\nabla e)$. Pairing with e, the periodic faces cancel and the lateral trace is zero. Hence

$$c^{n-1}\,\|e\|_V^2 \le -\langle \mathcal{R}(v), e\rangle \le \|\mathcal{R}(v)\|_{V^*}\,\|e\|_V.$$

Division, followed by Poincaré, proves the assertions. □

Theorem 12.2 (Barrier sup-norm certificates)**.** *Assume also $H_{\mathrm{fib}} r \succeq \kappa I$, $r < 0$ in the fiber, $r = 0$ on its boundary. Then*

$$|v-u| \le \frac{\|\mathcal{R}(v)\|_\infty}{n c^{n-1}\kappa}(-r). \tag{51}$$

If instead $0 < D_u, D_v \preceq CI$, the log-residual satisfies

$$|v-u| \le \frac{C\,\|\mathcal{L}(v)\|_\infty}{n\kappa}(-r). \tag{52}$$

Proof. For the first statement use the mean-cofactor operator $L = \overline{C} : H_{\mathrm{fib}}$, for which $Le = \mathcal{R}(v)$ and $Lr \ge nc^{n-1}\kappa$. Let the constant in front of $(-r)$ be b. Then $L(e + br) \ge 0$ and $L(e - br) \le 0$. Both have zero boundary values, so the maximum principle gives $e + br \le 0 \le e - br$. For the second statement use $\overline{B} = \int_0^1 D_{u+te}^{-1}\,\mathrm{d}t \succeq C^{-1}I$ and $\overline{B} : H_{\mathrm{fib}} e = \mathcal{L}(v)$. The same comparison applies. □

These are conditional estimates: a certified lower bound for the unknown exact Hessian is not provided by a sampled determinant alone. The lower bound on D_v cannot replace the bounds on both endpoints. For example, with $n = 2$, $r = |P|^2 - 1$, $B_0 = |P_1|^2/40$, $u = r/8$, and $v = r/4$, one

has $D_u = \mathrm{diag}(0.6, 0.5)$, $D_v = \mathrm{diag}(1.1, 1)$, and $\mathcal{R}(v) = 0.8$ when the prescribed determinant is 0.3. Using $c = 1$ from D_v alone and $\kappa = 4$ would falsely assert $\|v - u\|_\infty \le 0.1$, although the exact error is 0.125. An alternative a posteriori bracket avoids naming that unknown bound. If two explicitly constructed admissible functions $v + \delta r$ and $v - \delta r$ obey

$$\det_{\mathrm{M}} D_{v-\delta r} \le \lambda\mu \le \det_{\mathrm{M}} D_{v+\delta r} \quad \text{everywhere},$$

then comparison yields $v + \delta r \le u \le v - \delta r$. The inequalities must be continuum enclosures, not only values at collocation points.

Theorem 12.3 (Residual-to-energy gap)**.** *Use the normalized residual $R_v = \lambda^{-1} \det_{\mathrm{M}} D_v - \mu$. Suppose, along the segment from u to v, $\lambda^{-1}\mathcal{C}(D_{u+t(v-u)}) \succeq \kappa_0 I$. Then, for $r_* = \|R_v\|_{V^*}$,*

$$\|v - u\|_V \le r_*/\kappa_0, \qquad \frac{\kappa_0}{2}\|v - u\|_V^2 \le \mathcal{J}_z(u) - \mathcal{J}_z(v) \le \frac{r_*^2}{2\kappa_0}. \tag{53}$$

Proof. Equation (32) makes $\mathcal{J}_z$ strongly concave with modulus κ_0 along this segment. Its derivative at u is zero, while its derivative at v is R_v as a functional on V. Integrating the second variation gives the solution bound and the lower energy gap. Concavity at v gives, with $e = v - u$, $\mathcal{J}_z(u) - \mathcal{J}_z(v) \le r_*\|e\|_V - \kappa_0\|e\|_V^2/2 \le r_*^2/(2\kappa_0)$. □

An L^2 residual can be used through $\|R_v\|_{V^*} \le C_P\|R_v\|_2$. Approximate boundary data require a separately estimated lifting. The constants may deteriorate as $h \to 0$; none of these statements turns qualitative Dirichlet solvability into a uniform local-smoothing estimate.

13 A new off-solution energy identity

The most consequential error term is not just the norm of an elliptic residual on a single fiber. A trial potential also varies with the parameters z used in the energy propagation.

Theorem 13.1 (Exact determinant-energy defect)**.** *Keep the source's current A, geometry, density, and background exact. Fix $\lambda > 0$ independently of the parameter z, and retain the full moving-wall zero trace and common-null-line geometry of Section 8. Let v be a smooth real family of zero-trace potentials with $D_v = H_{\mathrm{fib}}(B_0 + v) > 0$. Define*

$$H(B_0 + v) = \sigma_v e_T e_T^* + \Sigma_v, \quad \Sigma_v \ge 0, \quad \mathcal{F}_v = \mathcal{J}_z(v), \quad R_v = \lambda^{-1}\det_{\mathrm{M}} D_v - \mu, \quad \rho_v = (v/r)|_{r=0}.$$

Then

$$\boxed{\Delta_z\mathcal{F}_v + \int_{\Omega_z} A : \Sigma_v \,\mathrm{d}\zeta = \int \rho_v G(r)^* A_{\mathrm{w}} G(r)\delta(r)\,\mathrm{d}\zeta + \int_{\Omega_z} R_v\sigma_v\,\mathrm{d}\zeta.} \tag{54}$$

Proof. The first Green calculation used in (30) applies to v and gives

$$\Delta_z \int (-v)\mu = \text{wall flux} - \int A : \Sigma_v - \int \mu\sigma_v.$$

The cofactor Green calculation still has zero wall term, independently of the determinant equation. Indeed, on the quaternionic tangent compression to $\nu^\perp$, zero trace gives

$$H(B_0 + \theta v)|_{\nu^\perp} = \big(H(B_0) + \theta\rho_v H(r)\big)|_{\nu^\perp}.$$

The shared null line makes this compression singular, so its normal cofactor vanishes. Integration of the cofactor contraction in θ then uses $\det_{\mathrm{M}} H(B_0) = 0$, since the full background has rank one. It gives $\Delta_z E_z(v) = \lambda^{-1} \int \det_{\mathrm{M}} H(B_0 + v)$. The Schur formula is $\lambda^{-1} \det_{\mathrm{M}} H(B_0 + v) = (\mu + R_v)\sigma_v$. Addition leaves exactly the final term in (54). □

Consequently

$$\left| \int R_v \sigma_v \right| \leq \|R_v\|_2 \, \|\sigma_v\|_2 \, .$$

A small determinant residual is insufficient without control of the spacetime Schur complement, which contains parameter derivatives and D_v^{-1}. Also $\mathcal{F}_v$ need not be nonnegative just because $D_v > 0$; the exact-solution comparison (33) cannot be reused for a trial field without justification. If only $|\mathcal{L}(v)| \leq \epsilon$ is certified, then $|R_v| \leq \mu(e^{\epsilon} - 1)$, retaining the actual density factor.

13.1 Additional current errors

Suppose a smooth real representation of a trial current has column divergence $b = \operatorname{div} A^{\mathbb{R}}$ rather than zero. In the total-domain Green identity an extra term appears:

$$\int_{\Omega} b \cdot (\eta \nabla v - v \nabla \eta). \tag{55}$$

To see this, take the divergence of $vA^{\mathbb{R}}\nabla\eta - \eta A^{\mathbb{R}}\nabla v$ and integrate. The cross gradient terms cancel by symmetry. This formula does not hide the derivatives on v or on the parameter test η. The sign in (55) is positive in the Green identity with left side $\int v A_{TT} \Delta_z \eta$. Since the energy uses $\int(-v)\mu$, its additional distributional contribution is the *negative* of (55). Errors in $A : H(B_0)$, tangency, or the boundary trace add their own terms. Preserving a positive Gram factorization alone does not make these defects vanish.

This provides a concrete target for a future Ram computation: certify the positive branch, full determinant residual, parameter derivatives, current conservation, and wall flux in norms compatible with (54). It is a stronger and more informative requirement than displaying a small residual at finitely many points.

14 Rank-one backgrounds: exact reduction and a solvable model

Proposition 14.1 (Rank-one determinant decomposition)**.** *For a quaternionic Hermitian H, a unit column e, and real λ,*

$$\det_{\mathrm{M}}(H + \lambda e e^*) = \det_{\mathrm{M}} H + \lambda \det_{\mathrm{M}}(H|_{e^{\perp}}). \tag{56}$$

The restriction means the compressed Hermitian form on the quaternionic orthogonal complement.

Proof. In a quaternionic unitary frame with first column e, write $H = \left(\begin{smallmatrix} a & b^* \\ b & D \end{smallmatrix}\right)$. When D is invertible, the Moore Schur formula gives $\det_{\mathrm{M}}(H + \lambda e e^*) = \det_{\mathrm{M}} D(a + \lambda - b^* D^{-1} b)$. Subtract the corresponding formula for H. Polynomial continuity in the entries extends the identity to singular D. □

In particular, dividing by $\lambda > 0$ separates a degree-$(n-1)$ reduced determinant from a correction $\lambda^{-1} \det_{\mathrm{M}} H$. If $\|H\|_{\mathrm{op}} \leq M$, the absolute correction is at most M^n/λ. This is a useful controlled asymptotic statement *only when the bound on H is available.* If $H = H_\lambda$ grows with λ, a large

background does not by itself make the correction small. For $n = 2$ the reduced determinant has degree one; for $n \geq 3$ it is generally still nonlinear. The source's planar alternative uses an exact mixed-cofactor energy argument, not merely an unjustified limit $\lambda \to \infty$.

Example 14.2 (Exact imaginary-ball solution)**.** On a unit ball in $P \in \mathbb{C}^n_{\mathbf{k}}$, with periodic real coordinates and no real-coordinate dependence, take $r = |P|^2 - 1$, $B_0 = \lambda|P_1|^2/4$, and constant $\mu > 0$. Then

$$u = \frac{\alpha}{4} r, \qquad D_u = \operatorname{diag}(\lambda + \alpha, \alpha, \dots, \alpha)$$

solves the zero-Dirichlet problem if and only if

$$\alpha^{n-1}(\lambda + \alpha) = \lambda\mu, \qquad \alpha > 0. \tag{57}$$

The left side is strictly increasing, so the positive root is unique. Let $d = n - 1$ and $\alpha_0 = \mu^{1/d}$. Then

$$\alpha_0(1 + \alpha_0/\lambda)^{-1/d} \leq \alpha \leq \alpha_0, \qquad 0 \leq \alpha_0 - \alpha \leq \frac{\alpha_0^2}{d\lambda}. \tag{58}$$

Indeed $\alpha = \alpha_0(1 + \alpha/\lambda)^{-1/d}$, and $1 - (1 + x)^{-1/d} \leq x/d$ for $x \geq 0$. A sharper a posteriori form is

$$0 \leq \alpha_0 - \alpha = \alpha\big((1 + \alpha/\lambda)^{1/d} - 1\big) \leq \frac{\alpha^2}{d\lambda},$$

by concavity of the $1/d$ power. For $n = 2$ the numerically stable closed form is $\alpha = 2\lambda\mu/(\sqrt{\lambda^2 + 4\lambda\mu} + \lambda)$. This model verifies the rank-one scaling and comparison directions. It is not the paper's moving current-generated fiber family.

15 The planar auxiliary equation: a genuine linear opportunity

Appendix D already supplies a different, linear construction for $n = 2$. At fixed parameters and passive coordinate Φ, put $B = H_{\mathrm{fib}}|\Phi|^2 = 4c^*c$ with $c = (\partial_P\Phi, \partial_Z\Phi)$. For $e = (c_2, -c_1)^{\mathsf{t}}$, $\mathcal{C}(B) = 4ee^*$. The operator is

$$L_B u = 4\sum_{i,j} \operatorname{Re}(e_i\bar{e}_j)\partial_{X_iX_j}u + 4|\mathfrak{J}|^2\Delta_W u, \qquad \mathfrak{J} = \det_{\mathbb{C}} \frac{\partial(W, \Phi)}{\partial(P, Z)}.$$

For each torus frequency k, its Dirichlet equation becomes

$$(\Delta_W - V_k)\widehat{u} = \frac{\widehat{\mu}}{4|\mathfrak{J}|^2}, \qquad V_k = \frac{|k \cdot e|^2}{|\mathfrak{J}|^2} \geq 0, \qquad \widehat{u}|_{|W|=w_0} = 0. \tag{59}$$

The zero Fourier mode is not a free constant: the disk Dirichlet Laplacian is invertible.

Theorem 15.1 (Planar inverse and extension of the forcing class)**.** *On the disk $D_R \subset \mathbb{R}^2$, let $V_k \geq 0$ be bounded for each fixed k and $A_k = -\Delta + V_k$ with Dirichlet form domain H^1_0. Equip H^1_0 with $\|\nabla w\|_2$ and H^{-1} with its dual norm. For $f_k \in H^{-1}(D_R)$ there is a unique weak solution to $A_k w_k = f_k$. For $f_k \in L^2$,*

$$\left\|A_k^{-1}\right\|_{L^2 \to L^2} \leq \lambda_1(D_R)^{-1}.$$

For any conforming approximation $\widetilde{w}_k$,

$$\|\nabla(w_k - \widetilde{w}_k)\|_2 \leq \|f_k - A_k\widetilde{w}_k\|_{H^{-1}}, \qquad \|w_k - \widetilde{w}_k\|_2 \leq \lambda_1^{-1}\|f_k - A_k\widetilde{w}_k\|_2. \tag{60}$$

The L^2 inverse extends modewise to square-summable divided forcing, with the same bound.

Proof. The form $a_k(w,w) = \|\nabla w\|_2^2 + \int V_k|w|^2$ is continuous and coercive on H_0^1. Lax–Milgram gives existence and uniqueness. Poincaré gives $a_k(w,w) \geq \lambda_1 \|w\|_2^2$. Apply these inequalities to the error equation and use duality. For infinitely many modes, Parseval and the uniform L^2 bound give convergence of the partial sums. Their equations hold weakly mode by mode; higher derivatives and parameter regularity require additional estimates. □

For the $k = 0$ constant-coefficient disk model, $\Delta u = f_0$ has $u(W) = f_0(|W|^2 - R^2)/4$. Thus the corresponding frozen $L_B u = \mu$ model has $u = \mu(|W|^2 - R^2)/(16|\mathfrak{J}|^2)$. One must not use this formula when $\mathfrak{J}$ actually varies with W. The source's energy $\mathcal{F}_{\mathrm{pl}} = \frac{1}{2}\int(-u)\mu$ and its mixed-cofactor identity are additional ingredients beyond solving (59).

15.1 A worked conforming polynomial calculation

On the unit disk take the manufactured model

$$A = -\Delta + 3, \quad u_*(r) = (1 - r^2)(1 + 0.4r^2 + 0.2r^4), \quad f = Au_*.$$

Use $\phi_j(r) = (1 - r^2)r^{2j}$ and assemble $K_{ij} = \int(\nabla\phi_i \cdot \nabla\phi_j + 3\phi_i\phi_j)$. These are boundary-conforming polynomial Ram panels, and also ordinary radial Galerkin polynomials. Writing $q = r^2$ makes every integral a polynomial integral on $[0, 1]$:

$$\int_{D_1} uv = \pi \int_0^1 u(q)v(q)\,\mathrm{d}q, \qquad \int_{D_1} \nabla u \cdot \nabla v = 4\pi \int_0^1 q\,u'(q)v'(q)\,\mathrm{d}q.$$

The tested errors are shown in Table 2. Since $A \succeq 3I$, the conservative certificate $\|u_* - \widetilde{u}\|_2 \leq \|f - A\widetilde{u}\|_2 /3$ applies. Enrichment improves this solvable model, but there is no demonstrated advantage over an equivalently assembled conventional Galerkin solver.

Highest j	L^2 error	L^2 residual	Residual bound/3	$\mathrm{cond}_2 K$
0	$1.10741\,10^{-1}$	6.49165	2.16388	1.000
1	$1.01778\,10^{-2}$	1.30713	0.435711	7.588
2	$1.92\,10^{-16}$	$6.41\,10^{-15}$	$2.14\,10^{-15}$	92.981

Table 2: Executed manufactured disk calculation. Integrals are evaluated from polynomial antiderivatives in floating point. The last row is roundoff-level exactness because u_* lies in the trial space.

16 Frames, propagation, and a quantitative reconstruction budget

16.1 TR1 provides an actual L^p interface

It would be too restrictive to say that the RMT corpus only has L^2 frame results. TR1-rmt-ext [16] constructs an L^p patch Banach frame using local inhomogeneous Haar systems and windows satisfying $\sum_j w_j^2 = 1$, with overlap at most M. Its coefficient norm is

$$\|c\|_{Y_p} = \left\| \left(\sum_j \sum_\ell |c_{j\ell}|^2 |h_{j\ell}|^2 \right)^{1/2} \right\|_p .$$

For $1 < p < \infty$, analysis and synthesis obey

$$c_{p,n} M^{-|1/p-1/2|} \|f\|_p \le \|\mathcal{A}f\|_{Y_p} \le C_{p,n} M^{|1/p-1/2|} \|f\|_p , \qquad \mathcal{D}\mathcal{A} = I.$$

The coefficient norm is not an unweighted ℓ^p norm. This removes an unjustified L^2-to-L^p identification but does not prove the time-evolution estimate: the matrix $\mathcal{A}U(T)\mathcal{D}$ still needs a frequency-uniform Y_p-to-spacetime bound. The geometric difficulty has been represented, not eliminated.

16.2 An unconditional error budget from L^2 data

Proposition 16.1 (Critical-norm cost of a band-limited reconstruction error)**.** *If $\widehat{e}_N$ is supported in a fixed annulus of radius N, then on a unit-length time interval*

$$\|U(T)e_N\|_{L^{p_c}_{T,x}} \le C N^{1/2} \|e_N\|_2 . \tag{61}$$

Proof. Bernstein at each time gives $\|U(T)e_N\|_{p_c} \le C N^{n(1/2-1/p_c)} \|U(T)e_N\|_2$. The exponent equals $1/2$, and the L^2 norm is conserved. Integrating over time proves the result. □

This does not assume the contested critical estimate. A sufficient budget to make reconstruction error negligible on its claimed scale is

$$\|e_N\|_2 = o\Big(N^{-1/2}[\log(2+N)]^{2/p_c} \|f_N\|_{p_c}\Big) .$$

One can insert TR248 or TR3 leakage, quadrature, and noise estimates into $\|e_N\|_2$, but a fixed relative L^2 error is not enough by itself. If the reconstruction is not annular, it must first be projected with the additional tails retained.

16.3 TR2: unitary propagation can still lose compressed observability

On a bounded frequency band let $H = |D|$, let V be an isometric finite trial synthesis, and let a contraction Q have leakage $\mathcal{E} = I - Q^*Q$. Define $K_T = V^* e^{\mathrm{i}TH} \mathcal{E} e^{-\mathrm{i}TH} V$. Exactly as in TR2's logarithmic-power commutator calculation,

$$(Qe^{-\mathrm{i}TH}V)^*(Qe^{-\mathrm{i}TH}V) = I - K_T, \qquad \|K_T - K_{T_0}\| \le |T - T_0| \, \|[H, \mathcal{E}]\| . \tag{62}$$

Differentiate the bounded exponentials and integrate to prove the inequality. Thus $\|K_{T_0}\| + |T - T_0| \, \|[H, \mathcal{E}]\| < 1$ is a sufficient observation window with a positive lower gain. This is a direct wave-specific adaptation of TR2's method, not an identity between H and $\log A$. The exact Gram may be better than this conservative bound. No information is lost by the uncompressed unitary wave; the loss is caused by compression after phase evolution.

16.4 A window wave is not automatically the same wave

For two real symbols $\omega, \widetilde{\omega}$ on an N-annulus, put $d_N = \sup |\widetilde{\omega} - \omega|$. Plancherel gives

$$\Big\| (e^{-\mathrm{i}T\widetilde{\omega}(D)} - e^{-\mathrm{i}T\omega(D)}) f_N \Big\|_2 \le \min(2, |T| d_N) \, \|f_N\|_2 . \tag{63}$$

For the one-dimensional window-wave example in the MathWorld supplement [26], $\omega_b(\xi) = c|\xi|\{1 - (b\xi)^2/40 + O((b\xi)^4)\}$. At fixed nonzero time, small phase error in this regime needs $b^2 N^3 \to 0$, not merely $bN \to 0$. A sufficient scaling is $b = o(N^{-3/2})$. A fixed-window saturated generator is a different high-frequency equation. This warning does not rule out useful nonlocal wave models; it prevents them from being used as evidence about the wrong propagator.

17 Where the logarithms arise and what cannot yet be improved

The source proves, on a cap and its fiber family,

$$\mathcal{F}(0, |\tau| = 4a) \lesssim a^2 \, \|f\|_{p_c}^{p_c}, \tag{64}$$

$$\mathcal{F}(T, |\tau| = 2a) \gtrsim a^2 \, \|U(T)f\|_{p_c}^{p_c}, \tag{65}$$

$$\Delta_z \mathcal{F} \le V(s)\mathcal{F}, \qquad V(s) = Cs^{-2}e^{-cs/h}. \tag{66}$$

These uniform bounds, not the mere ability to solve an elliptic PDE, are decisive. On the wall the Gaussian decays as $e^{-c_0 s/h}$. Near the center, inversion costs $e^{C\gamma^2 s/h}$. Choosing the cap aperture γ small before fixing K preserves a net decaying exponential.

The lower comparison uses a ball of radius h in the $3n$ spatial slots and a density floor $m_X \gtrsim (hs)^{-d}\sigma^{-m}e^{-C\gamma^2\sigma}|U(T)f(X)|^2$, where $\sigma = s/h$. The quadratic barrier has depth $\gtrsim h^2 m_X^{1/d}$. Multiplying by the density yields $m_X^{1+1/d}$ and hence the wave power $2(1 + 1/d) = p_c$. The spatial convolution ensures the density floor throughout the ball, not merely at its center.

17.1 The logarithmic radial operator

For the circular mean of a nonnegative supersolution, the radial comparison equation is $u'' + s^{-1}u' = (V + \kappa^2)u$. Set $x = \log(s/b)$; then

$$u_{xx} = s^2(V + \kappa^2)u.$$

The natural quantity controlling the comparison is $\int_b^3 s \log(s/b)V(s)\,\mathrm{d}s$. For the potential in (66) this is bounded independently of h. The outward passage from the inner wall to a fixed-radius band costs one logarithm. The return passage to radius $2b$ costs another. The resulting estimate is

$$H(0, 2b) \gtrsim [\log(3/b)]^{-2} \int_1^2 H(T, b)\,\mathrm{d}T.$$

Substituting the two bounds in (66), canceling a^2, and taking p_c-th roots gives the exponent $2/p_c$.

17.2 A scalar obstruction to removing one logarithm

The source provides a concrete obstruction, which can be checked directly. Let $D = \log(3/b)$,

$$p(s) = \frac{\log(3/s)}{D}, \quad g(s) = \frac{s^2(1 + \log(3/s))}{4D}, \quad L(s) = p(s)g(b) + (1 - p(s))g(3).$$

Then $\Delta_\tau g = p$ and $L - g \ge 0$ by the maximum principle. The function $E(T, s) = (1 - \cos T)p(s) + L(s) - g(s)$ is nonnegative and $\Delta_{T,\tau}E = (\cos T - 1)p(s) \le 0$. Its inner trace is $1 - \cos T$, while

$$E(0, 2b) \le \frac{9 \log 2}{4D^2}.$$

Thus a better scalar comparison for *all* such supersolutions is impossible. Ram moments or a different basis for the same radial ODE cannot remove this counterexample. Any one-logarithm improvement must use additional structure enjoyed by actual wave energies but not by arbitrary scalar supersolutions.

17.3 Sharpness and restricted improvements already in the source

A radial frequency-annular wave focusing at $T = 3/2$ has, on its outgoing shell, size comparable to $N^{(n+1)/2} r^{-d/2}$ and thickness N^{-1}. At $p = p_c$, integration of its p-th power contributes $\int_{A/N}^{1/4} r^{-1}\,\mathrm{d}r$. This forces the power $1/p_c$. A separate packet construction obstructs a lossless weak spacetime estimate for $2 < p \le p_c$. Neither obstruction is defeated by an invertible coordinate change.

Appendix B treats angular regularity. For angular degree at most M, it asserts the stronger critical bound with factor $[\log(2+N)]^{1/p_c} + [\log(2+M)]^{2/p_c}$. For unrestricted angular data an angular logarithmic norm with power $B > 1/2 + 1/p_c$ is sufficient in its theorem. These are real restricted improvements, but they are already contributions of the source, not new Ram results.

18 Transfer, consequences, and extension boundaries

The source covers the annulus by a fixed number of small angular caps. Its torus-to-Euclidean transfer uses integration by parts outside the wave cone to control the far kernel and bounded-overlap physical localization for the near part. A number of caps growing with N would change the loss unless additional orthogonality were proved.

Interpolation with L^2 conservation and the annular L^∞ kernel estimate yields the finite-scale factor

$$N^{s_+(p)}[\log(2+N)]^{\beta(p)}, \qquad \beta(p) = \min\{d(1-2/p),\, 2/p\}.$$

Summation with an arbitrarily small Sobolev loss gives (2). The source also gives logarithmic Sobolev refinements through sequence Hölder and Littlewood–Paley theory. Those summation bounds are not automatically optimal in every already-known high-p range.

Its downstream claims include the following implications:

1. A modulated time average selects a thin spherical frequency shell. Focusing the corresponding input yields a local spherical extension estimate with the same logarithmic power.
2. Random signs on disjoint angular caps turn the expected squared amplitude into a sum of mode energies. This is a justified averaging step in the restriction-to-tube argument, unlike discarding coherent terms in the deterministic current (17). The tube-overlap estimate carries twice the logarithmic power.
3. The Kakeya maximal bound must hold for arbitrary tube centers and separated directions. The Hausdorff conclusion additionally uses variable-scale covers, not only one-scale neighborhood volume.
4. Thin radial-shell decomposition gives Bochner–Riesz bounds at strict orders. The Stein square-function argument combines radius scales inside a Hilbert norm and retains its vector-valued estimate.
5. Parabolic rescaling of spherical caps and the Fresnel formula give paraboloid and Schrödinger consequences. Phase errors, dilation Jacobians, and epsilon-removal hypotheses are part of the implication.

All these are source-dependent in this report. No additional general Kakeya, restriction, Bochner–Riesz, or Schrödinger conjecture is claimed to have been proved by a Ram representation.

Appendix E already extends the source mechanism to a simple largest eigenvalue $\phi(\xi)$ of an $n \times n$ real symmetric linear pencil in n spatial variables, with positive transverse curvature on a specified

conic patch. The polarization map is then a local angular coordinate, and a positive quadratic form supplies the Gaussian. The gap, extremal sign, curvature, and matching state dimension matter. Crossing eigenvalues, nonextremal sheets, arbitrary matrix sizes, or variable-coefficient backgrounds require a new argument.

19 Executed checks and what they establish

The new reproducibility script `local-smoothing-support/check_extensions.py` uses NumPy 2.3.5 and random seed 20261026. Its recorded output is `local-smoothing-support/extension-check-results.json`. These are deterministic floating-point algebra and manufactured-solution checks, not interval proofs, a nonlinear PDE computation on the source fibers, or a benchmark against the source's analysis.

Check	Cases	Maximum observed discrepancy
Positive quaternionic rank-one determinant identity	120	$1.79\,10^{-15}$ relative
Moore determinant directional derivative	120	$3.63\,10^{-10}$ scaled
Quaternionic Schur identity	120	$9.98\,10^{-16}$ relative
Imaginary-shift / directional Gaussian identity	500	$3.34\,10^{-16}$ absolute
Fixed-κ Schrödinger equation	24 modes	$6.99\,10^{-16}$
Covariant window heat equation	24 modes	zero to reported precision
Positive-root ball determinant model	9	$8.89\,10^{-16}$ relative
Mixed-slice quaternion commutator	100	$7.03\,10^{-16}$ block norm
Common right-phase invariance of the current	100	$8.40\,10^{-15}$ block norm

Table 3: Finite checks of identities proved in the report. Directional derivatives use Richardson-corrected central differences; their larger error is consistent with floating-point differentiation.

Quaternionic matrices were represented by the faithful complex blocks

$$a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} \longmapsto \begin{pmatrix} a + \mathrm{i}b & c + \mathrm{i}d \\ -c + \mathrm{i}d & a - \mathrm{i}b \end{pmatrix}.$$

For *positive* Hermitian inputs the Moore determinant was computed as the positive square root of the complex determinant. This procedure was not used to infer the signed Moore determinant of indefinite matrices. The finite derivative tests supplement, rather than replace, the all-input proof of (19).

The added TR3 checks preserve quaternion multiplication order in these same complex blocks. In 100 random cases the current-error bound in (45) held, with maximum observed left/right ratio 0.863189. The two projected imaginary channels had the exact Gram matrix $\mathrm{diag}(1, 1, 2)$. Common right-unit phases left the Gram current unchanged to roundoff, despite changing its quaternionic factor. Block Frobenius discrepancies in Table 3 use the complex representation; the tested physical current bound uses quaternionic norms, obtained by dividing squared block norms by two. These checks support implementation consistency, not a new observability theorem for the full nonlinear PDE.

The spectral Taylor checks gave:

δ	m	Observed maximum error	Proved bound
0.5	4	$2.40174\,10^{-4}$	$2.60417\,10^{-4}$
1	6	$1.76115\,10^{-4}$	$1.98413\,10^{-4}$
2	10	$4.39055\,10^{-5}$	$5.13067\,10^{-5}$

The observed maxima are over 1001 scalar sample points; the continuum guarantee comes from Theorem 6.1, not from the sampling. For the ball model with $n = 3, \mu = 2, \lambda = 100$, $\alpha = 1.4043864384351816$ and $\alpha_0 - \alpha = 0.00982712394 < 0.01$, as (58) predicts.

A deliberate negative control is $\mathrm{diag}(-1, -1, 1)$: it has determinant one but a negative eigenvalue. Thus even an exact positive determinant residual does not imply positive definiteness. The conforming polynomial disk example in Table 2 demonstrates convergence on its named model and increasing matrix conditioning cost, not an across-problem computational advantage.

20 Comparison: benefits, costs, and attribution

Table 4: Conventional/source description versus the Ram-based reformulation used here.

Object	Ram formulation and concrete benefit	Limitation or cost
Imaginary light-ray continuation	Nonnegative spectral generator and finite moment remainder; explicit inverse condition number on a band	Backward inversion remains unstable outside the controlled band; critical L^p derivatives are separate.
Quaternionic wave current	Ordered real blocks, mixed-slice commutator bound, physical metric and channel leakage	No permission to reorder factors; positivity, tangency, and divergence must all be retained.
Nonlinear determinant fiber PDE	Full cofactor Jacobian, conforming Ram panels, admissibility-aware residual and energy certificates	No generic moment inverse; nonlinear convergence, continuum positivity, and h-uniformity remain to be proved.
Planar disk inverse	Boundary-adapted polynomial/Green representation and extension to square-summable weak forcing	This linear reduction is already in the source; equal Galerkin spaces have equal mathematical approximation power.
Finite frame measurements	TR1's specified L^p space; TR3's ordered constrained Gram; TR2's propagation leakage margin	A frame is not a local-smoothing inequality; compressed unitary evolution may lose observability.
Approximate proof energies	Exact residual–Schur defect and full-divergence test term	Small pointwise or sampled elliptic residual alone does not control propagation.
Critical logarithmic loss	Logarithmic radial coordinates identify exactly where the two factors enter	The scalar counterexample survives every equivalent coordinate representation.

20.1 What is proved here

The independent results include the bandwise Gaussian remainder and inverse budget, the covariant window equations, the mixed-slice order-error bound, the conforming cofactor stiffness inequalities, the residual and energy-gap certificates, the off-solution energy identity, the controlled rank-one

model, the weak planar inverse extension, and the frequency/phase reconstruction budgets. Their proofs identify the assumptions under which they are valid. They are additions to this application-level synthesis, not claims that Taylor expansion, spectral calculus, Lax–Milgram, Schur complements, or Galerkin theory originated with Ram Calculus.

20.2 What is specific to the Ram organization

The corpus supplies a consistent interface between window measurements, ordered hypercomplex channels, moment derivatives, physical frame metrics, boundary constraints, and nonlinear residuals. Using it here exposes errors that would be easy to hide in an informal symbolic inversion: missing side order, ignored normalization derivatives, false off-range roots, omitted tails, or absent Dirichlet closure. This is a useful contribution to transparency and reproducibility.

The established mathematical bounds are also available to a conventional analyst who supplies the same hypotheses and performs the same calculations. No experiment here demonstrates a unique Ram accuracy, runtime, memory, or asymptotic-complexity advantage. That distinction preserves the value of the exact reformulations without overstating their originality or performance.

21 Research directions with testable success criteria

1. **A complete planar proof-computation benchmark.** Implement the actual W-dependent disk operators, not only a frozen model. Certify zero traces, spatial Fourier tails, parameter derivatives, and wall flux. A success criterion is a continuum bound on the right side of (54) at both comparison circles.
2. **Scaled positive-branch validation for $n \geq 3$.** Use the source's barriers as sub- and supersolutions, with interval or analytic Hessian bounds. Track cofactor coercivity and Schur factors as functions of h. A fixed-h converged Newton solve is a first test, not the required uniform theorem.
3. **Coherent quaternionic compression.** Use TR3 channels with the correct packed Hermitian metric, preserve the common-right-phase gauge, and certify (45). Measure divergence and tangency defects separately. A small Gram error alone is not enough.
4. **A wave-specific one-logarithm mechanism.** Search for additional identities or restrictions on actual $\mathcal{F}$ that rule out the scalar counterexample in Section 17. Angular regularity is an existing example of extra information, not a new unrestricted theorem.
5. **True L^p coefficient propagation.** Apply TR1's Banach-frame norm to the wave evolution matrix and seek uniform bounds that use cone geometry. The success criterion is the full scale-independent input quantifier, not a sampled matrix norm at a fixed cutoff.
6. **Rough coefficients and curved sheets.** Treat coefficients, spectral gaps, curvature, and frame derivatives as explicit defect terms and compare with the hypotheses of current rough-wave literature. Avoid extrapolating Appendix E beyond its simple extremal sheet assumptions.
7. **Residual-driven adaptive panels with a certified tail.** TR4's resolved–tail Schur test can potentially select a smaller effective model, but must certify the ambient inverse and nonlinear radius. Compare work and error with conventional spectral and finite element solvers on the same physical problem.

None of these proposed improvements is claimed to have been completed by the finite checks in this report.

22 Conclusion

Ram Calculus is substantively relevant to the source paper. The most direct connections are the exact Gaussian/moment representation, the corrected quaternionic real-block calculus of TR60, the constrained channel framework of TR3, and the boundary-aware linearizations of the auxiliary Moore-determinant equation. TR1, TR2, and TR4 add useful L^p reconstruction, phase-leakage, and nonlinear validation tools.

The auxiliary PDE is nevertheless not automatically inverted by the Ram partial differential operator. Its positive cone, real scalar potential, moving boundary, rank-one background, and conserved current are indispensable. The proved certificates and off-solution identity provide a precise way to apply Ram methods while measuring what an approximation changes. They offer a disciplined route to further analysis and validated computation. They do not yet improve the unrestricted exponent or close the gap between the source's $2/p_c$ upper logarithm and the $1/p_c$ lower obstruction.

A Local source crosswalk and reproducibility

All paths in this appendix are local reference locations, not public download links. Let D_0 denote `C:/Users/mural/OneDrive/Documents/` and let T_0 denote D_0 followed by `MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/`. The following sources were used for the named mathematical roles:

Identifier	Relative path and role
TR60 original	T_0+`TR60-Quaternionic-RMT/Quaternionic_RMT.tex`; historical comparison, not the default authority for corrected inverse claims.
TR060 revised	D_0+`RamFoundationTRs/TR-rev-1-old-tr060/TR-rev-1-old-tr060.tex`; sided definitions, real blocks, weak transpose, conditional frames.
TR062 revised	D_0+`RamFoundationTRs/TR-rev-1-old-tr062/TR-rev-1-old-tr062.tex`; kernel moments, full jet rows, finite inverse hypotheses.
Operator foundations	D_0+`RamOperator/RamOperator-Foundations.tex`; revised translation, window, and inverse convergence conditions. The associated `RamOperatorSeries-rev2.tex` and `RamOperatorTR2-rev2.tex` were also used.
RMT extensions 1–4	D_0+`RamRMT-extensions/`; `TR1-rmt-ext-frames-atoms.tex`, `TR2-rmt-ext-fractional-complex.tex`, `TR3-rmt-ext-hypercomplex-constraints.tex`, `TR4-rmt-ext-nonlinear.tex`.
TR216	T_0+`TR216-microlocal-pseudodifferential-ram-calculus/TR216-microlocal-pseudodifferential-ram-calculus.tex`; full differentiated tests and boundary remainders.
TR248	T_0+`TR248-Stable-Adaptive-Ram-Frames/TR248-Stable-Adaptive-Ram-Frames.tex`; leakage, noise, physical metric, and conditioning.
TR250	T_0+`TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex`; compatible weak forms and trace obligations.
TR205 and item 293	T_0+`TR205-openai/v4-fable51/sections/p8.tex`; D_0+`MyDocuments2026/res-rt/res-math/TR-item-293-astra.tex`; determinant residuals and branch limitations.

Identifier	Relative path and role
Volume 12	D_0+`RamCalBooks-v2/Volume12-Multidimensional-Transform-Geometry/source/nie8/`; especially `ch03.tex`, `ch04.tex`, `ch10.tex`.
Wave foundations	D_0+`RamCalBooks-v2/Volume13-PDE-Weak-Solutions/source/nie9/ch03.tex` and `ch04.tex`; classical wave kernels and energy domains.
Supplement	D_0+`mathworld-supplement/Ram-Calculus-MathWorld-Supplement-v5-final-draft.tex`; finite-window dispersion and limits.
Prior OpenAI screen	D_0+`OpenAI-722-math/00-Applicability-Catalogue/OpenAI-722-Ram-Applicability.tex`; earlier bounded screening of different wave/restriction manuscripts.

The new output and supporting checks are under D_0+`res-papers/OpenAI-papers/`. The public source snapshot is `local-smoothing-support/source-42907c9/`. Its section files provide exact source locators: `current.tex`, `calculus.tex`, `energy.tex`, `barrier.tex`, `propagation.tex`, `transfer.tex`, `sharpness.tex`, `kakeya.tex`, `consequences.tex`, `stein_square.tex`, `paraboloic.tex`, `dirichlet.tex`, `angular.tex`, `weak.tex`, `planar.tex`, and `pencil.tex`. The local audit notes record the bounded checks and their limitations. Existing research sources were not edited.

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